DEVELOPMENT OF A WEBSITE LEARNING MANAGEMENT SYSTEM FOR PHILIPPINE AGRICULTURAL ENGINEERING STANDARDS

A Capstone Project Proposal
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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

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Introduction

Project Context

Education plays an important role in human development. It imparts knowledge, skills, talents, and values that are essential for individual and societal growth/development. Sharma and Ankit (2023) highlighted that education is among the most concrete sources of information and knowledge. A quality education may assist individuals in reaching their life objectives making them globally competent and capable. In the field of Agricultural and Biosystems Engineering, education is crucial as it equips individuals and future professionals to use appropriate cutting-edge technology, sustainable practices, and efficient natural resource management to address societal issues, particularly in the agricultural industry. However, education and the process of learning might be challenging, especially when reviewing complex concepts.

The Philippine Agricultural and Engineering Standards (PAES) are a technical standard for agricultural engineering in the Philippines that significantly improves the technical performance of agricultural engineering projects. It emphasizes sound agricultural engineering principles and practices required in the planning, design, construction, operation, and optimization of agricultural engineering technologies, systems, and services to efficiently address problems related to agricultural engineering technologies, such as irrigation, postharvest, structures, rural electrification, mechanization, alternative energy, domestic water supply, farm roads, and bridges. Learning and familiarizing these standards are crucial for ABE students to make future designs and endeavors safe, efficient, and successful. Thus, PAES is highly incorporated in their academic curriculum as it is one of the most important foundations of the course. However, the content of PAES is highly technical, making it difficult to understand crucial ideas. Moreover, the vast and extensive concepts presented can be overwhelming, resulting in students' loss of enthusiasm in the learning process and difficulty absorbing and retaining relevant contents efficiently.

Traditional method of reviewing PAES involves reading the standards and presenting it in

classes, and is often inefficient and unengaging for students. Meanwhile, passive learning is ineffective for knowledge retention and does not promote active student participation. The enormous content of PAES is highly technical which poses challenges for students. As a result, many students fail to do well on academic assessments, limiting their professional growth in the field of ABE.

Thus, the project intends to solve the limitations of current PAES review practices by creating a User-Centric Website Learning System. This system will utilize technology and modern instructional design concepts to provide ABE students with an engaging and effective learning experience. The website, which incorporates components and multimedia materials, will offer students a dynamic platform for reviewing PAES at their own pace and convenience. The project intends to give an efficient learning platform for students that will cater to their respective needs and preferences.

The project has the potential to improve the learning experience of ABE students which can result in efficient PAES review and enhance learning outcomes.The website's interface will help students learn complicated ideas and remember technical knowledge under PAES. This will result in greater academic success and better preparation for the demands of the ABE field. Furthermore, the study will add to the existing body of knowledge about instructional design and technology-enhanced learning, with significance for various subjects and disciplines.

Purpose and Description

The purpose of this capstone project is to develop an online Learning Management System (LMS) for Agricultural and Biosystems Engineering (ABE) students in the Philippines. The goal is to make learning the Philippine Agricultural and Engineering Standards (PAES) easier and more engaging.

Traditional ways of studying PAES involve a lot of reading and lectures, which can be overwhelming. Our project changes this by turning these review methods into an online

format. Using modern technology, the Learning Management System helps students learn PAES at their own speed and convenience.

The LMS will include videos, learning materials, quizzes, tutorials, discussion forums, virtual activities, and progress tracking. These tools make learning more interesting and help students understand complex concepts better. The platform offers a personalized learning experience that suits different learning styles and preferences.

With a user-friendly interface, flexible learning options, and technical support, the LMS allows students to study in a way that fits their needs. By breaking down the material into smaller parts, the LMS helps keep students interested and improves their understanding of PAES.

Objectives

The objectives of the study is the following:

- To develop a web-based platform for ABE students to review Philippine Agricultural Engineering Standards (PAES);
- To design a user-friendly interface so students can easily navigate and review PAES at their own pace;
- To incorporate videos and materials to make learning more engaging;
- To include quizzes and exercises to help students check their understanding of PAES.
- To create a learning environment that fits different learning styles and preferences.
- To develop a system that will be evaluated by 50 students, 5 teachers, and 3 IT experts.

Scope and Limitations

The purpose of this project is to build a web-based system to help Agricultural and Biosystems Engineering (ABE) students review the Philippine Agricultural and Engineering Standards (PAES). The primary users of this system will be ABE students and teachers. The project is scheduled to be developed over the academic year. Key features of the system will include videos, quizzes, exercises, and an easy-to-use interface to enhance the learning experience.

However, there are certain limitations to consider. The system's effectiveness may be limited by the availability and reliability of internet access. While the system will cover PNS/BAFS 308: 2021 – Agricultural Structure – Lairage Swine, PAES 301:2000 – Engineering Materials – V-belts and Pulleys for Agricultural Machines – Specifications and Applications, PNS/BAFS/PAES 216:2017 – General Irrigation Terminologies, PAES 101:2000 – Agricultural Machinery – Technical Means for Ensuring Safety – Guidelines (Old), and PAES 203:2000 Agricultural Machinery – Moisture Content Determination for Rice and Corn, it may not encompass every detail due to the extensive amount of information. The success of the system also depends on the active participation and engagement of the students. Additionally, the project may face limitations due to time, budget, and technical resources, which could impact the scope and quality of the final product.

Related Literature

Philippine Agricultural Engineering Standards (PAES)

PAES are critical guidelines that ensure the quality and safety of agricultural practices and equipment in the Philippines. Implementing these standards in an educational context ensures that future agricultural engineers are well-versed in national regulations and best practices. According to the Philippine Society of Agricultural Engineers (PSAE), PAES covers a wide range of topics, from the design and construction of agricultural structures to the maintenance of machinery and equipment.

The standards included in PAES were developed through the efforts and cooperation of various government agencies, private organizations, and academic institutions. These standards are continuously updated and improved to keep up with the fast pace of development in various fields of science and technology.

A web-based learning and assessment system to support flexible education

The implementation of web-based learning and assessment systems in higher education has garnered significant attention due to the challenges posed by increasing costs and resource constraints. White et al. (Year) present a comprehensive examination of a web-based learning and assessment system known as the Computer-supported Education and Instructional Learning (CECIL) system. Designed to alleviate administrative burdens and enhance educational flexibility, CECIL offers a multifaceted approach to teaching, learning, and resource management.

The CECIL system embodies the potential benefits of integrating technology into educational practices. By leveraging web-based platforms, universities can optimize resource allocation, streamline administrative processes, and foster collaborative learning environments. Through its web-based interface, CECIL facilitates access to learning

materials, assessment tools, and administrative functionalities, thereby enhancing the overall educational experience for both students and instructors.

This research contributes to the discourse on web-based learning management systems by highlighting the transformative potential of such systems in addressing the evolving needs of higher education. By embracing technology-driven solutions like CECIL, institutions can adapt to changing pedagogical paradigms and promote educational innovation in the face of resource constraints.

Academic and student use of a learning management system: Implications for quality

The study by Concannon, Flynn, and Campbell (2005) investigates how Learning Management Systems (LMS) are used in higher education and their implications for educational quality. This research specifically examines the WebCT system at a large university in Australia, focusing on the experiences and needs of both students and faculty.

The study reveals that student satisfaction with the LMS is largely dependent on the integration of the system into the teaching process. Factors such as well-organized courses, ample resources, timely feedback, and effective communication with instructors were crucial for a positive student experience. In contrast, faculty members were more concerned with technical and regulatory challenges related to LMS usage, rather than pedagogical considerations.

Learning Management System (LMS) Impact on Student Engagement and Learning Outcomes

The integration of Learning Management Systems (LMS) into higher education institutions has revolutionized the traditional classroom setting by providing a digital environment for educational interactions. As highlighted by Nor Adzharuddin et al. (Year), LMS platforms serve as dynamic tools for managing learning interventions and delivering educational content to students. With the widespread adoption of LMS across universities worldwide, it becomes imperative to explore the feedback and experiences of

students as primary users of these systems.

Nor Adzharuddin et al. delve into the phenomenon of LMS adoption among university students, aiming to provide valuable insights into its efficacy and impact on higher learning environments. By synthesizing previous studies and empirical findings, this paper offers a comprehensive overview of the diverse perspectives surrounding LMS implementation. The research underscores the multifaceted nature of LMS usage in higher education, acknowledging both its potential benefits and challenges. Through an examination of student feedback and perceptions, Nor Adzharuddin et al. shed light on the acceptance and effectiveness of LMS platforms in facilitating learning experiences. This exploration of LMS dynamics provides valuable insights for educators, administrators, and developers seeking to optimize the use of technology in educational settings.

Adopting Web-Based Learning and Teaching: A Case Study in Higher Education

In the evolution of education, the integration of learning and teaching on the web has become more common. Samarawickrema and Stacey (2007) discuss the transformative potential of web-based learning and teaching in their study. They highlight the main departure from traditional classrooms, providing teachers and students with a dynamic platform to publish and find information. This study combines existing research and insights to shed light on the education revolution, highlighting both its benefits and challenges.

Understanding Faculty Use of the Learning Management System

The Learning Management System (LMS) plays a crucial role in higher education, facilitating both instructors' and students' participation. According to a report by the Educause Center for Research and Inquiry in 2014, nearly all universities (99%) utilize an LMS, with a significant proportion of instructors (85%) and students (83%) actively using it. This indicates a widespread adoption of LMS in educational settings, marking a significant shift from its previous status of being underutilized and distinct.

The study by West, Waddoups, & Graham (2006) examines how a large university in the

Midwest has been utilizing the same online learning framework for 15 years. By analyzing system logs and database queries, the study investigates which features of the LMS are most utilized and how their usage evolves over time. Furthermore, it compares the usage of LMS in traditional classes versus fully online classes to discern any differences in utilization patterns.

Web-based Learning: Characteristics, Practices, Challenges and Recommendations

The study "Web-based learning: Characteristics, Practices, Challenges and Recommendations" looks at how web-based learning affects online education. Explore the basics of designing online learning, its advantages and difficulties. Also, learn about the best ways to do it and hear what students and teachers have to say. This study shows how important it is to keep up with new technology in schools. It shows how important it is for teachers to know how to use digital technology in order to teach well online. This shows that organizations should focus on helping people learn how to use computers and the internet so they can do well in online classes.

The research supports using constructivist methods in teaching and recommends that online learning should concentrate on creating tools for developing the skills needed for modern education. About your project called "E-Learning Management System for Philippine Agricultural Engineering Standards", this study gives important information about the features, difficulties, and suggestions for online learning. Because your project is about online learning and systems on the internet, the findings and suggestions from this study can help you create and use your learning system. By following the study's advice and using the best methods, you can make sure that your online learning system works well for the Philippine Agricultural Technical Standards and takes advantage of the newest technology for web-based learning.

User interface aspects of a Web-based educational system

The review, titled "A Client Interface Point of view of an Internet Counseling System," looks at how shrewd frameworks and online learning stages are interfacing. Propels in computer-based learning and the developing utilization of the Web have driven to the

improvement of online learning frameworks that are getting to be exceptionally critical. This article examines the significance of making adaptable online learning frameworks that can adjust and provide valuable data. In brief, this upgrade makes the composing instrument indeed way better by including portable and web highlights, making the instruction framework simpler to utilize.

This idea is noteworthy in making a difference to move forward the “Web-Based Learning Administration System for Provincial Building Courses within the Philippines”. "Both think about the significance of online stages for instruction. Consolidating shrewdly highlights into the client interface of an instructive framework can move forward learning, as the learning administration framework proposed to utilize Web innovation in provincial zones. Clients with numerous ways to connect, instructive frameworks can offer assistance understudies learn way better by adjusting to their needs and utilize modern advances.

Research looks at how intelligent computer systems can be used in conjunction with online learning sites. As technology advances and more people use the Internet for education, it is even more important to create flexible systems for online education. The goal of the research is to make the educational system easier to use by adding mobile and web features to the interface. This research is important for "Web-based Learning Management System for Provincial Engineering Courses in the Philippines". Both studies illustrate the importance of online platforms for education. Adding smart features to school computer systems can improve learning, especially in rural areas. The new learning system will help students with the Internet. By providing students with different learning styles, they receive a personalized education that can change based on their needs and incorporate new technologies.

The Impact of Online Quizzes on Student Engagement and Learning3

The challenges of fostering active engagement in educational settings have long been noted as troublesome for both instructors and students. The purpose of the Teaching, Learning, and Innovation (TLI) grant was to encourage greater student interest and active

participation in learning while ensuring meaningful engagement in class discussions and group activities. Drawing on principles from educational research, the initiative hypothesized that providing students with online reading quizzes would motivate them to engage with course materials, participate more actively in discussions, and ultimately perform better on exams.

In the fall of 2011, both sections of PSYCH 270 implemented this approach, utilizing the quizzes a total of 25 times. Researchers anticipated that students would find the online quizzes enjoyable and perceive them as valuable tools for learning. Designed to reinforce knowledge incrementally throughout the semester rather than relying on last-minute cramming before exams, the quizzes aimed to enhance the learning process.

The study sought to examine the influence of online quizzes on student engagement and learning, particularly in large classes. Findings suggest that the inclusion of online quizzes encourages students to engage more actively in lessons, collaborate on projects, and participate in meaningful discussions. These results provide insight into the role of online assessments in promoting active learning and fostering a more interactive educational environment in larger lecture-based courses.

ONLINE QUIZZES IN A VIRTUAL LEARNING ENVIRONMENT AS A TOOL FOR FORMATIVE ASSESSMENT

A study examined the use of online quizzes in virtual classrooms to determine their effectiveness in improving undergraduate learning outcomes in college settings. The research focused on 204 undergraduate students studying life sciences who were enrolled in an introductory physics course. The course utilized a combination of face-to-face lectures, group activities, and the online learning platform Moodle. Weekly assessments were conducted through both written and online quizzes administered via Moodle.

The study evaluated students' performance on online quizzes and their perceptions of these quizzes compared to traditional written tests, using a survey as a key tool for data collection. Findings revealed that both written and online quizzes served as strong

predictors of students' performance on the final exam. Additionally, students' performance on the quizzes showed improvement over time. Students reported a preference for online quizzes, noting that they found them engaging and beneficial to their academic studies. These results suggest that online quizzes can play a significant role in enhancing learning experiences and outcomes in higher education.

Online Discussion Forum as a Tool for Interactive Learning and Communication

Interactive instructing in online dialog forums The rise of advanced stages has revolutionized the way millennials learn and communicate. These stages encourage exceedingly intelligently learning situations backed by expanding web entrance. The study "Online Talk Gatherings as Apparatuses for Intelligently Learning and Communication" highlights the significance of online talk gatherings (ODFs) in upgrading student-centered instructional approaches at all instructive levels. ODF, as well as the “CSC Forum " created for the Office of Computer Science, Tai Solarin College of Instruction, Nigeria, are imperative instruments for advancing intuitively learning and moving forward communication between understudies and instructors (Adebayo, 2019). It upgrades communication and learning

The CSC Gathering is planned to encourage the trade of thoughts among understudies with respect to course substance and assignments, while making strides in the quality of communication and connections between understudies and workforce. Highlights such as the Live Chat module empower dynamic discussions and bunch talks, making it simple for understudies to take part in real-time discussions. This gathering not as it were advances scholastic discourse, but moreover reinforces the department's data proficiency program and progresses students' computerized proficiency aptitudes. This thought highlights the significance of instructors giving direction in online gatherings to invigorate understudy intrigued and engagement (Adebayo, 2019). Preferences and challenges of online discourse forums

ODF has demonstrated itself in upgrading interaction between understudies and instructors, communication and dispersal of data in almost school and office exercises. Be

that as it may, they consider moreover distinguished a few issues that will restrain the victory of online gatherings. These issues incorporate deficient control, moo understudy and educator engagement, destitute quality entries, hostile entries, deferred criticism, destitute web network and time limitations. In spite of these impediments, the potential benefits of ODF in making intelligence and locks in learning situations stay considerable (Adebayo, 2019).

Learning Analytics for Tracking Student Progress in LMS

Learning analytics includes the collection, investigation, and announcement of information about learners and their settings, with the point of understanding and optimizing learning and the situations in which it happens. The speeding up of computerized change in instruction, driven by the COVID-19 widespread, has opened up the significance of successful learning analytics, especially within the setting of Learning Administration Frameworks (LMS) (Siemens, 2013).Role of LMS in Present day Education Learning Administration Frameworks have ended up necessarily to advanced instruction by giving a stage to oversee, provide, and track instructive substance and understudy advance. LMS stages offer a range of highlights that encourage and upgrade the e-learning encounter, counting course organization, substance conveyance, and evaluation devices (Watson & Watson, 2007). In any case, a significant crevice remains within the capacity of these frameworks to successfully track and analyze students' directions of progress.

The paper titled “Learning Analytics for Following Understudy Advance in LMS” addresses this hole by leveraging learning analytics to track understudy advancements. The investigation highlights that LMS produces significant sums of information through understudy intuitive with the stage. Analyzing this information can surrender important experiences into students' advance, making a difference teachers distinguish lower entertainers, anticipate understudies at hazard, and decide those in require of extra bolster (Ferguson, 2012).Key Discoveries and Implementation The case consider conducted at Cadi Ayyad College included following students' advance in a Web advances course

utilizing characterized Key Execution Markers (KPIs). The discoveries uncovered that whereas the markers were viable in observing advance, they were less fruitful in assessing in general understudy execution. This underscores the requirement for more comprehensive analytics instruments that not as it were track advance but too give an all encompassing see of understudy execution (Greller & Drachsler, 2012).

Integrating learning analytics into LMS can essentially upgrade instructive results by providing timely and noteworthy bits of knowledge into understudy execution. This will offer assistance teachers tailor meditations to meet desires of personal learners, in this manner making strides in victory rates and the general viability of the learning handle. The review proposes that ceaseless refinement of analytics instruments and markers is fundamental to completely realize the potential of learning analytics in instruction (Sclater, 2017).

Literature Review: Learning Management Systems in Higher Education

The discussion titled "Writing Audit: LEARNING Administration Frameworks" digs into the noteworthy contemplation that higher education educators confront when actualizing a Learning Administration Framework (LMS). One of the first contemplations is the monetary toll related to conveying an LMS. Sellers regularly offer comprehensive LMS items that incorporate different highlights to upgrade online instruction, but these items come with significant forthright costs and yearly location permit expenses. Such charges can be especially expensive for littler sections or education with restricted resources.

This conversation is so important for the "E-BASED LEARNING Organization System FOR PHILIPPINE Agrarian Planning Benchmarks" because it focuses on the financial and logistical hurdles of putting an LMS into action. Understanding these challenges is crucial for making a cost-effective and viable LMS custom fitted to the specific needs of Philippine rustic building instruction. By recognizing the money related restrictions, the wander can explore elective subsidizing sources, cost-sharing models, or more sensible LMS options to ensure productive utilization without compromising quality.

This dialog highlights the importance of choosing an LMS that balances strengths with thoughtful consideration. It is often particularly important to create a learning management system for rural design measures, as specialized content and functionality is required. The experience of writing this review can lead to the identification and customization of an LMS to meet the educational needs of rural design students while staying within budget requirements.

Implementing Web-Based Learning and Teaching Using Mobile Technology

The rise of portable innovation in education, In later decades, portable innovation has gotten to be a necessary portion of lifestyle due to its availability and broad accessibility. This has produced solid intrigue from analysts and education in joining portable gadgets into learning situations. The use of portable innovation in instruction points to extend understudy inspiration, make strides student-teacher interaction, and eventually give distant better;a much better;a higher;a stronger;an improved">a stronger learning involvement (Partner, 2009). The benefits of portable learning Mobile instruction offers numerous points of interest such as more noteworthy availability, adaptability and comfort. It permits understudies to get to instructive substance and assets anytime, anyplace, which is particularly helpful in today's fast-paced world. The compactness of portable gadgets bolsters distinctive learning styles and empowers maintained engagement with learning materials, subsequently expanding maintenance and understudy accomplishment (Kukulska-Hulme & Traxler, 2005). Case Ponder: A College Portable Learning System

This chapter of the book "Actualizing Web-based Learning and Educating with Versatile Advances" centers on the improvement and execution of a versatile learning framework outlined to encourage intuitively learning in a college environment. The framework gives understudies with versatile get to to different highlights to upgrade the learning handle, counting evaluation and input instruments. Case thinks about demonstrate that the framework has been tried and assessed with palatable comes about, proposing that it is compelling in upgrading the learning encounter (Adeoye & Wentling, 2017). The usage

of this system increases understudy inspiration, moves forward interaction and increments the productivity of the learning prepare. These discoveries highlight the potential of versatile innovations to convert instructive hone by making learning more energetic and open (Adeoye & Wentling, 2017). Significance to Agrarian and Biosystems Designing Education Applying this ponder to the improvement of a web-based learning administration framework (LMS) for rural and biosystems building within the Philippines might altogether move forward the learning involvement. Joining versatile innovation into an LMS can increment openness and engagement, making complex specialized substance more sensible for understudies. This approach is steady with the objective of utilizing innovation to back compelling learning and instructing in agrarian designing (Elefan, 2006).

EVALUATION MODEL FOR THE USABILITY OF WEB-BASED LEARNING MANAGEMENT SYSTEMS WITH USER PROFILE

Presentations to Web-Based Instructive Assets Administration Frameworks (WERMS) Web-Based Instructive Assets Administration Frameworks (WERMS) are outlined to provide information and share data to bolster learners successfully and effectively through progressed electronic advances. These frameworks play a pivotal part in present day instruction by encouraging learning exercises and upgrading the by and large instructive encounter (Alonso et al., 2005). In any case, the ease of use of these frameworks remains a noteworthy challenge in their plan and implementation. Usability within the setting of HCI alludes to the degree to which a system enables its clients to realize their objectives successfully, proficiently, and palatably. Key ease of use properties incorporate adequacy, productivity, fulfillment, learnability, interactivity, consistency, inspiration, and client control. These qualities are basic for making WERMS that give a consistent relationship between client data and framework substance (Nielsen, 1994).

The ponder titled “Evaluation Show for the Convenience of Web-Based Learning Administration Frameworks with Client Profile” proposes a comprehensive show for surveying the ease of use of WERMS. The demonstration joins different convenience

properties such as adequacy, productivity, fulfillment, learnability, inspiration, interactivity, consistency, learner control, and client profile (age, sex, and level of encounter). These traits were assessed through surveys and tests to decide their effect on the convenience of WERMS (Smith & Jones, 2021).Findings and Implications The think about found that all the proposed qualities altogether influence the convenience of WERMS. This shows that planning user-centric WERMS requires conscious thought of these variables to guarantee an ideal client encounter. The assessment demonstrates an organized approach to surveying and progressing the convenience of WERMS, which is pivotal for upgrading learning results and client fulfillment (Smith & Jones, 2021).

Relevance to Agrarian and Biosystems Designing Education

In the setting of creating a web-based Learning Management Framework (LMS) for Rural and Biosystems Designing within the Philippines, joining the discoveries from this ponder is crucial. By guaranteeing the LMS is planned with a solid center on convenience, teachers can improve the learning encounter, making complex specialized substances more available and locked in for understudies. This approach adjusts with the objective of moving forward instructive adequacy and effectiveness through progressed electronic advances (Elefan, 2006).

Web-Based Learning Environment: A Theory-Based Design Process for Development and Evaluation

The study "Web learning environments: a theoretical design process for development and evaluation" deals with the design and evaluation of user-oriented web learning support environments with a specific focus on Global Positioning System (GPS) education. . . In response to the increasing growth of online courses and programs, this study addresses two key weaknesses of existing approaches: the integration of user interface design with instructional design, and the development of an evaluation framework to improve the overall quality of the online learning environment.

This study uses the Integrated Design Process (IDP) framework, which includes a theoretical approach to improving the design process and usability of distributed and

distributed learning environments (WD2L). Using the IDP, this study aims to assess the equivalence of the WD2L environment to traditional supplementary learning and to assess users' perceptions of its resources. The results demonstrate the effectiveness of the proposed IDP in improving the WD2L environment and highlight its equivalence with traditional supplementary learning, especially as a web-based supplementary education program.

Users have expressed positive opinions about WD2L's eco-friendly resources. The study highlights the importance of considering all aspects of the learning environment, including applied domain knowledge, conceptual learning theory, instructional design, user interface design, and overall quality assessment. By addressing these aspects, this study contributes to the understanding of effective design processes and evaluation frameworks for web-based learning environments.

Comparison of Websites and Mobile Applications for E-Learning

Information and communication innovation (ICT) has significantly changed the instructive environment and gives assorted and advancing apparatuses for e-learning. With the quick advancement of smartphone computer programs and equipment, unused conceivable outcomes have developed to improve the e-learning encounter. The company's objective is to form productive and user-friendly applications that give the finest client involvement, particularly within the field of instruction (Selim, 2007). Comparison of web and portable applications in e-learning

The Consider "Comparison of e-learning websites and portable applications" explored client inclinations for e-learning applications on web and versatile stages. This consideration is part of a larger venture that examines the potential of e-learning in higher instruction utilizing different e-learning applications. The primary stage of this ponder included two bunches of clients aged 21-30 who connected with a portable application and a web application containing data approximately worldwide understudies at an American college. Criticism was collected through semi-structured interviews and it was found that users' encounters and encounters with ICT applications essentially affected

their discernments of the application (Mahmood & Qureshi, 2020). Client inclinations and application efficiency.

Research has found that the sort of data and its representation play a basic part in deciding the adequacy and convenience of web and portable applications. Clients appreciate the availability and comfort of portable apps, but they too appreciate the breadth and structure of web apps. These discoveries propose that both stages have one of a kind focal points and can complement each other to supply a capable e-learning environment (Dabbagh & Kitsantas, 2012). Suggestions for higher education.

The come about of this consider have imperative suggestions for end of e-learning in higher instruction. Understanding client inclinations and the one of a kind qualities of web and portable applications can offer assistance to teach plan more successful e-learning arrangements. The following stage of this investigation will investigate this with a more extensive group of onlookers, which can give more experiences into optimizing e-learning stages to meet distinctive client needs (Mahmood & Qureshi, 2020). Significance to Rural and Biosystems Building Education

Applying these discoveries to the improvement of a web-based learning administration framework (LMS) for agrarian and biosystems building within the Philippines can upgrade the learning encounter by leveraging web and versatile advances. This approach is steady with the objective of making complex specialized substances more available and locked in for understudies, subsequently making strides learning results and planning them for proficient challenges in rural building.

The focus of this study centers on the use of computers and the internet in agricultural education, highlighting the importance of integrating electronic learning into agricultural instruction. With the continuous advancement of farming technologies and knowledge, there is a pressing global need for improved agricultural education systems. The research showcases the development of an e-Learning platform, "eLearn Agriculture," by the Indian Agricultural Statistics Research Institute (IASRI) in New Delhi. This system was

tested with sample classes in Agricultural Statistics and Computer Applications, featuring structured topics, interactive activities, videos, quizzes, and assignments.

The "eLearn Agriculture" platform is designed with distinct functionalities for educators, students, evaluators, and system administrators. It incorporates tools for creating mathematical equations and embedding multimedia content such as videos into courses. The platform leverages the open-source MOODLE system to develop and manage online learning materials and users. MOODLE's modular design allows for easy integration of features such as equation editors, mathematical symbols, and user-friendly interfaces, enhancing the system's functionality.

This platform enables agricultural educators to publish course materials online, offering a standard framework for course creation and delivery. The study provides insights that are valuable for the development of an e-Learning Management System tailored to the Philippine Agricultural Engineering Standards. Both this study and similar research emphasize that incorporating technology into education enhances student learning and facilitates more flexible and interactive learning methods. The "eLearn Agriculture" system exemplifies how Learning Management Systems (LMS) can be effectively integrated into agricultural education, offering practical strategies for creating web-based learning platforms to improve education in agricultural engineering.

A mobile and web-based student learning system

Introduction to portable and web-based learning systems Mobile and web-based learning frameworks have become fundamental instruments in advanced instruction, giving adaptable and intuitive stages for understudies to lock in with learning materials. These frameworks utilize progressed innovation to create instructive assets more open and locked in, subsequently upgrading the learning involvement (Partner, 2009).

The integration of instructive innovation, particularly through versatile and web-based frameworks, bolsters diverse learning styles and gives a range of intelligent instruments. These apparatuses incorporate instructional exercises, works out, intuitive movements,

appraisal apparatuses, chat rooms, and educational games. These highlights meet the wants of understudies by giving distinctive ways of connection with and understanding the fabric (Sharples et al. , 2007). Case think about: the correct utilization of the vitality learning system.

The book entitled "Portable and Web-based Understudy Learning Framework" depicts a framework planned to educate tall school understudies at the Setubal Organized of Innovation / Setubal Organized of Innovation approximately the judicious utilization of vitality. The framework is outlined to teach understudies on the significance of vitality proficiency and vitality preservation since the power era depends on fossil powers. By combining different educational instruments and intuitive components, the framework moves forward understudying engagement and learning results (Pereira & Costa, 2012).

A Dedicated Web-Based Learning System

Given the title "Dedicated Web-based Learning System" and the abstract provided, this study presents a prototype learning-oriented e-learning system designed to address the limitations of traditional Learning Management Systems (LMS) and Learning Content Management Systems (LCMS).).). Recognizing the frustration teachers and students feel with current learning management systems, he emphasizes the need for an innovative and easy-to-use alternative that empowers students and instructors. The paper highlights the challenges faced by students, particularly students from developing countries and the Middle East, including limited English proficiency and computer skills, which are barriers to designing and implementing e-learning systems. To this end, the proposed prototype aims to simplify the learning process so that even those with intermediate English and general computer skills can navigate the system and access study materials.

This consideration proposes a computer designing course plan that regards SCORM benchmarks within the proposed framework. It emphasizes the significance of understudy criticism in refining the model and prescribes ceaseless change based on examination. With respect to your extent titled “E-Learning Administration Framework for Philippine Rural Designing Standards”, this considers the advancement of a specialized e-learning

framework that's reasonable for the requirements of understudies with distinctive dialect aptitudes and computer abilities. Indeed in case your extension focuses on a diverse area of education, the standards and strategies depicted within the consideration can assist you plan and actualize your learning administration framework.

Related Studies and/or Systems

Microsoft Learn

Microsoft Learn is an online learning platform developed and delivered by Microsoft that offers a variety of courses and resources based on Microsoft products, as well as certification. Microsoft Learn provides examinations, hands-on labs, and interactive modules to help users gain knowledge and skills in a variety of subjects, including cloud computing, data science, and software development. Microsoft Learn's integration and structured learning paths with Microsoft certification make it an extremely beneficial resource for professionals looking to advance their knowledge of Microsoft technology.

STI eLMS

STI E-Learning Management System (eLMS) is a LMS developed by the STI Education Systems Group. It is designed to provide blended learning and enhance the educational experience of the students and instructors at STI College campuses. It features course content management, grade tracking, online assessment, and communication tools. The STI eLMS is specifically made for the needs of STI’s curriculum by providing a cohesive and integrated learning environment.

Google Classroom

Google Classroom is a cloud-based learning management system (LMS) built for seamless integration with other Google services. It assists instructors in creating compelling learning experiences by seamlessly and intuitively integrating Google technologies like Google Drive and Google Docs, as well as allowing users to personalize, control, and measure their education.

CVSU e-Learning

The Cavite State University (CVSU) e-Learning platform is an in-house learning management system designed specifically for CVSU's numerous academic programs. It provides course management, multimedia material distribution, evaluation tools, and communication channels. It utilizes the Moodle platform to offer integrated and blended learning possibilities to CVSU students and teachers.

Synthesis

The review of related literature and studies provides insights into the design, implementation, and evaluation of web-based learning systems. Key studies emphasize integrating user interface design with instructional design and developing robust evaluation frameworks to improve the quality of web-based learning environments. The study "Web-Based Learning Environment: A Theory-Based Design Process for Development and Evaluation" emphasizes the need for a user-centered approach and comprehensive evaluation to enhance the learning experience. "A Dedicated Web-Based Learning System" underscores the need for a learner-oriented system accommodating users with varying levels of English proficiency and computer skills.

Research on online quizzes shows their effectiveness in enhancing student engagement and learning outcomes. The study "Academic and Student Use of a Learning Management System: Implications for Quality" explores disparities between students and attributes expectations and experiences with Learning Management Systems, emphasizing the importance of well-designed courses, timely feedback, and effective instructor interaction for student satisfaction. Faculty responses focus on technical and administrative challenges, indicating a need for quality control and comprehensive support.

These thoughts collectively emphasize the significance of well-designed, user-centered web-based learning situations that consolidate both guidelines and client interface plan standards. They too highlight the potential of online developmental survey devices, such as tests, to upgrade understudy engagement and learning results. For the proposed

project, "Web-Based Learning Management System for Philippine Agricultural Engineering Standards," these experiences are necessary. The essential points to create a specialized LMS that not as it were coordinating high-quality substance but to support possible educating objectives and give a user-friendly implication. Given the challenges and openings distinguished within the looked into writing, the proposed extend is justified in its point to make a sturdy, adaptable, and learner-oriented framework that addresses the particular needs of Agricultural Engineering Standards within the Philippines. By centering on quality control, comprehensive back, and the integration of academically sound sharp, the project has the potential to essentially improve the learning encounter and instructive results for both understudies and staff.

TECHNICAL BACKGROUND

Overview of Current Technologies to be Used in the System

1. Front-End Technologies

HTML5: The latest version of the HyperText Markup Language, used to structure and present content on the web, offering improved multimedia support, semantic elements, and cross-platform compatibility.

CSS3: The latest version of Cascading Style Sheets, used to style and format web pages with advanced features like animations, gradients, and responsive designs for modern web development.

JavaScript: A versatile, high-level programming language used to create interactive and dynamic content on web pages, such as animations, form validation, and real-time updates.

Bootstrap 5: A popular front-end framework for building responsive, mobile-first websites and web applications, featuring pre-designed components, a grid system, and utility classes for faster development.

2. Back-End Technologies

PHP: A widely-used open-source scripting language suited for web development. PHP is known for its efficiency in server-side scripting, and it will be utilized for dynamic content generation, form handling, and session management.

3. Database Technologies

MySQL: A relational database management system used for structured data, offering reliable transaction support and ease of integration with other systems.

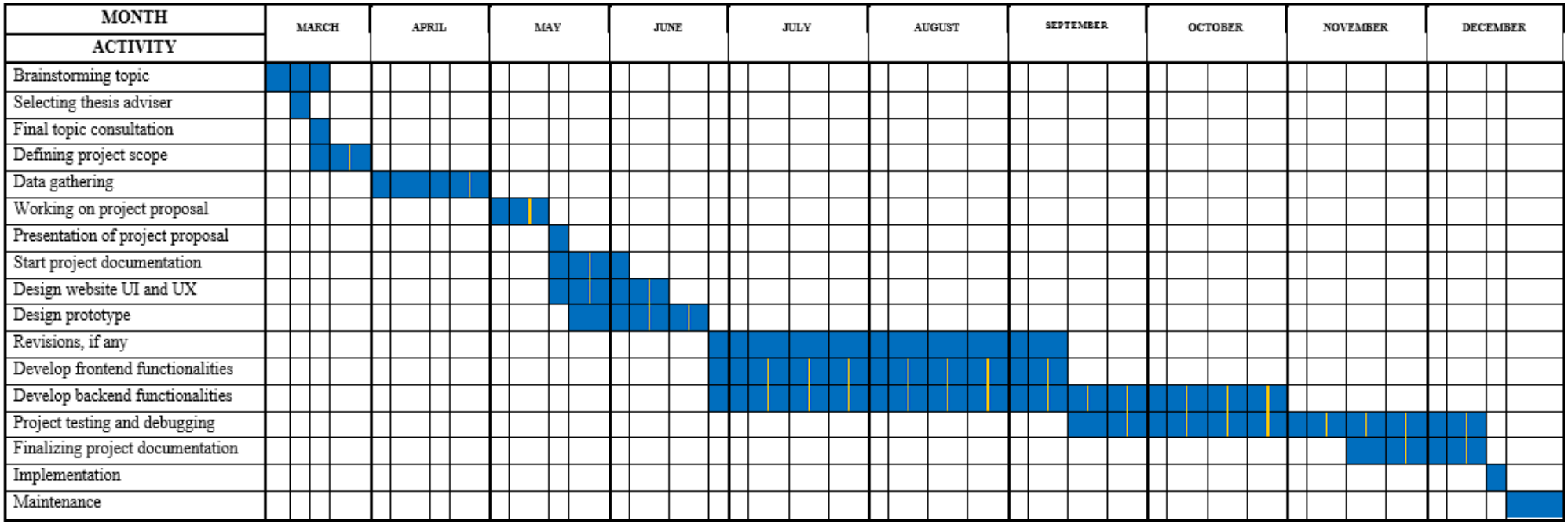
4. Development Tools and Platforms

Git and GitHub: Version control system and repository hosting service for collaborative development and source code management.

XAMPP: A lightweight, preconfigured software package that provides a local web server environment with Apache, PHP, MariaDB, and Perl, making it easy to develop and test web applications.

Visual Studio Code: A lightweight, open-source code editor with built-in support for multiple programming languages, extensions, and debugging tools, making it ideal for developers to write, test, and manage code efficiently.

Gantt Chart of Activities



Resources

- Hardware

  Laptop - with the following specifications:

  - Processor: Intel Core i3 and up
  - RAM: 4gb
  - Storage: 120GB SSD/HDD
  - Operating System: Windows 7 to latest

- Software

  Integrated Development Environment (IDE)

  - Visual Studio Code

  Version Control System

  - GitHub

METHODOLOGY

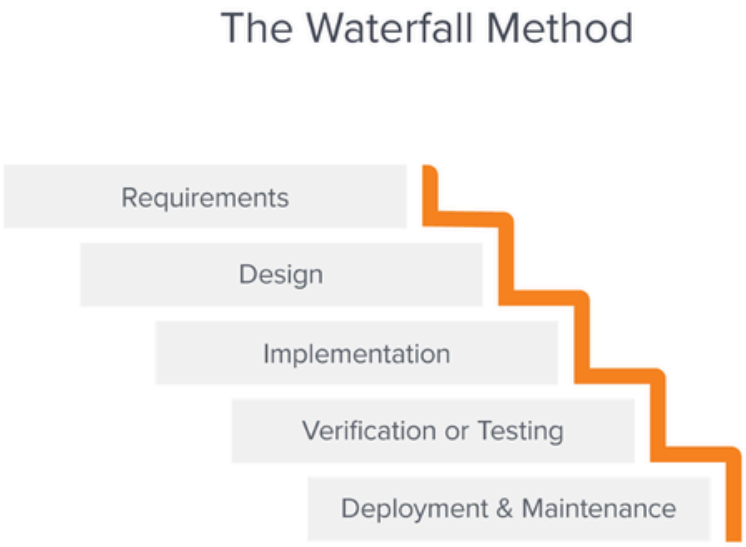


Figure: 1 Waterfall Method SDLC

The researchers chose the Software Development Life Cycle (SDLC) waterfall model for its structured and sequential approach, ensuring systematic progress through each phase of software development without overlapping, which is crucial for managing complexity and maintaining clarity in our project.

Requirements: The initial needs of a capstone project, such as creating an LMS, are frequently steady and well-defined. Such projects are well suited for the waterfall method, which begins with requirements gathering and moves through phases in a systematic manner. This makes it possible to fully comprehend and record the goals that the LMS must accomplish, guaranteeing clarity right away.

Design: The waterfall technique places a strong emphasis on a methodical approach in which every stage builds on the one before it. The design phase concentrates on developing comprehensive specifications and architectural blueprints for the LMS after requirements have

been established. By doing this, you can be sure that every part of the system has been carefully thought out before it is put into use, which lowers the possibility of significant redesigns or modifications later on in the project.

Implementation: the system undergoes rigorous testing to ensure it aligns with specified requirements and operates seamlessly. This process is particularly critical for an LMS, where reliability and functionality are paramount, as it enables the identification and resolution of any potential issues prior to deployment.

Verification and testing: play a crucial role in the waterfall model, especially in the context of an LMS. During the testing phase, which occurs after the system has been fully implemented, rigorous testing is conducted to ensure that the LMS functions as intended and meets all specified requirements. This phase is essential for identifying and rectifying any issues or discrepancies before the system is deployed to end-users.

Deployment: the waterfall model transitions into the maintenance phase. **Maintenance** involves ongoing support and updates to the LMS to ensure its continued functionality, security, and alignment with evolving educational needs. This phase may include bug fixes, performance enhancements, and the incorporation of new features based on user feedback and technological advancements.

Requirements Analysis

Based on the survey conducted, the expectations and needs for the proposed Web-based Learning System for Philippine Agricultural Engineering Standards (PAES) have been outlined. The study identified key aspects such as the target audience, the system's intended purpose, its usage context, and the means through which users will access the platform.

The survey gathered responses from a diverse group within the Agricultural and Biosystems Engineering (ABE) community, including students from various year levels, as well as faculty members and graduates, who represented a smaller portion of the respondents. The primary objective of the system is to assist students and instructors in understanding PAES by providing

streamlined access to educational materials and tools. Challenges faced by students in learning PAES include a lack of illustrative examples, complex language, limited study time, difficulty in locating resources, and information overload.

Respondents emphasized the importance of PAES in the curriculum and its relevance to professional practice in the field. Desired features for the system include an intuitive interface, learning materials, progress tracking, discussion forums, and compatibility with mobile devices. Students prefer learning resources that incorporate a mix of text, quizzes, graphics, and sample calculations.

The survey revealed strong agreement on the need for a specialized PAES platform, with students likely to adopt it if it offers simplicity, up-to-date content, mobile compatibility, and is endorsed by a trusted organization. Technical support was also identified as a critical component. Laptops and smartphones were identified as the preferred devices for accessing the platform.

In conclusion, the survey highlights a clear demand for a user-friendly, web-based learning system for PAES. Such a platform should provide comprehensive, engaging content and flexible accessibility to enhance learning experiences for students and professionals in the ABE field.

Requirements Documentation

The Learning Management System (LMS) is specifically designed to meet the needs of Agricultural and Biosystems Engineering (ABE) students by offering a comprehensive platform to facilitate the study of the Philippine Agricultural Engineering Standards (PAES). Its goal is to provide an engaging and effective learning experience while assisting instructors in delivering structured materials and tracking student performance.

The LMS takes inspiration from Microsoft Learn and incorporates features that address the challenges faced by ABE students. The platform provides PAES-related educational content through innovative tools that encourage active participation, including quizzes and multimedia resources. By integrating a variety of teaching methods, the system helps students develop a deeper understanding of PAES topics.

Students and faculty can benefit from tools that allow them to visualize progress within the course. These tools ensure a clear understanding of where students stand in their studies, making it easier for instructors to identify areas requiring further attention.

The LMS promotes collaboration and peer interaction through integrated platforms for discussion and knowledge sharing. These collaborative tools help create a sense of community among students and provide opportunities to exchange insights and solve problems collectively.

With its mobile-responsive design, the LMS ensures flexibility, allowing students and instructors to access the system anytime and anywhere. This feature provides a seamless experience across devices, ensuring that users can engage with the platform without technological barriers.

By incorporating user feedback from ABE students and instructors, the LMS offers a targeted solution for the effective study and application of PAES, enhancing accessibility, interactivity, and the overall quality of education.

Design of Software, System, Product, and/or Processes

In designing the Learning Management System (LMS) for Agricultural and Biosystems Engineering (ABE) students, faculty, and graduates, a meticulous and standards-compliant approach will be implemented to ensure functionality, ease of use, and adherence to relevant guidelines.

The LMS will feature a minimalistic and streamlined user interface (UI) to promote intuitive navigation and accessibility for all users. A modular design approach will be adopted, with the main dashboard serving as a central hub. This hub will provide quick access to learning materials, reviewers, resources, and progress tracking tools, ensuring that users can efficiently locate and interact with the platform's features.

To align with Philippine educational standards, all content and instructional designs will be validated in collaboration with experts in PAES and ABE. Comprehensive modules will focus on PAES-related topics, incorporating text-based materials, components, quizzes, and potentially video tutorials. These modules will be carefully designed to match learning objectives outlined

by the Philippine Agricultural Engineering Standards, ensuring their applicability and relevance.

The primary focus of the design will be to create a system that is straightforward and user-friendly for students, faculty, and other users. Continuous feedback collection from users will guide iterative improvements to the interface and functionality. The LMS will also feature a responsive design, guaranteeing compatibility across a range of devices, including desktops, smartphones, and laptops. Adaptive layouts and scalable elements will provide a seamless user experience, regardless of screen size.

Security and privacy will be top priorities, with the LMS adhering to the Data Privacy Act of the Philippines and other relevant regulations. Measures such as secure login mechanisms and regular audits will be employed to protect user data. Additionally, the system will be engineered to handle substantial workloads, ensuring responsiveness and reliability under heavy usage.

To cater to a diverse user base, the LMS will include language options for both English and Filipino, allowing users to choose their preferred language for easier comprehension of the content. Repeated testing and feedback loops will be integral to the development process, ensuring that each feature aligns with user needs and system requirements.

By incorporating validated content, a robust design, and adherence to both Philippine and LMS standards, this platform aims to significantly enhance the learning experience for ABE students, faculty, and graduates. The LMS will serve as an effective and reliable tool for studying and applying the Philippine Agricultural Engineering Standards.

Development

The development process of the Learning Management System (LMS) was guided by established educational and software development standards to ensure a functional, user-friendly, and effective platform tailored for Agricultural and Biosystems Engineering (ABE) students, faculty, and graduates.c

1. Development Framework and Tools

The LMS was built using modern web technologies and frameworks that support modularity, scalability, and responsiveness:

- Backend Development: Implemented using MySQL to handle server-side logic, database interactions, and APIs.
- Frontend Development: Developed with HTML5, CSS3, Bootstrap5, and JavaScript frameworks for creating an intuitive and responsive user interface.
- Database Management: Relational database systems like MySQL were used to store user data, course materials, and progress tracking securely.

2. Standards Compliance

The LMS adheres to local standards:

- Philippine Educational Standards: Course content and instructional materials were validated with experts in PAES and ABE to ensure alignment with the Philippine Agricultural Engineering Standards.
- Data Privacy and Security Standards: The system complies with the Data Privacy Act of the Philippines by implementing secure login mechanisms, data encryption, and user access controls.

3. Modular Design Approach

The LMS was developed using a modular design to provide flexibility and scalability. Key modules include:

- Dashboard Module: Centralized hub for accessing course materials, resources, and progress tracking.
- Course Management Module: Allows instructors to upload materials, create quizzes.
- Progress Tracking Module: Enables users to monitor completion.
- Responsive Design Module: Ensures accessibility across devices, achieved through CSS frameworks like Bootstrap.

4. Iterative Development Process

The development team followed an iterative process based on the Agile methodology:

- Planning: Defined system requirements and objectives based on user feedback and standards.
- Design: Created prototypes and wireframes for user interface and experience testing.
- Development: Implemented functionalities in sprints, testing each module for usability and compliance.
- Testing: Conducted unit testing, integration testing, and user acceptance testing to ensure system reliability and functionality.

5. Deployment

The LMS was deployed on a reliable hosting platform, ensuring high availability and accessibility. Regular updates and maintenance are planned to keep the system secure and up-to-date.

RESULTS AND DISCUSSION

Testing

The LMS was developed and tested in compliance with established software development standards to ensure reliability, security, and user satisfaction. Key testing processes included:

1. Functional Testing: Verified that each module performed as intended, including course management, quizzes, and progress tracking.
2. Integration Testing: Assessed seamless interaction among modules, such as data flow from course materials to progress tracking.
3. Usability Testing: Evaluated the user experience, ensuring the interface was intuitive for students and instructors.
4. Security Testing: Validated that the system adhered to the Data Privacy Act of the Philippines, protecting user data through secure login, encryption, and access controls.

Key Outcomes

1. All modules passed functional and integration tests.
2. The system demonstrated high responsiveness
3. User feedback from usability tests indicated satisfaction with the interface and overall functionality.

Description of Prototype

System Requirements

Hardware Requirements:

- User Devices: Desktop, laptop, or smartphone with a modern browser.

Software Requirements:

- XAMPP as the development and testing platform.

- Backend: PHP and MySQL.
- Frontend: HTML5, CSS3, JavaScript, Bootstrap 5.

Network Requirements:

- Stable internet connection for accessing the LMS.

Preliminary Design

Dashboard: A user-friendly interface acting as the central hub for accessing all LMS features, such as course materials, quizzes, and progress tracking.

Modules:

- Course Management: For admin to upload materials and quizzes.
- Progress Tracking: Allows students and instructors to monitor course progress.
- Responsive Design: Ensures accessibility on devices of all sizes.

Evaluation and Testing

- Alpha Testing: Conducted by developers to identify and resolve initial issues.
- Beta Testing: Involved students and instructors to gather real-world feedback.

Implementation Plan

Overview

The Learning Management System (LMS) will transition from the development phase to full operational use, with the website scheduled for public release around **February 2025**. This plan outlines the steps necessary to ensure the successful deployment, installation, and transition of the LMS into a fully functional educational platform.

Major Tasks

1. **Pre-Deployment Preparation:**

- Finalize the testing phase and address any outstanding issues.
- Upload all LMS files to the server and configure the database for live deployment.

- Conduct final system testing in the live environment to ensure readiness.

2. **Deployment:**
   - Publish the LMS website by **February 2025**, making it accessible to all intended users.
   - Provide login credentials to students, Instructors, and administrators.
   - Release onboarding instructions and guides for initial users.
3. **Training:**
   - Provide a comprehensive user manual for students, detailing how to navigate the platform and use its features.
4. **Post-Deployment Support:**
   - Establish a technical support team to address user queries and troubleshoot issues during the early adoption phase.
   - Plan regular updates and maintenance to improve system performance and security over time.

Resources Needed

- **Hardware:** A production-grade server with adequate storage, memory, and processing power to host the LMS.
- **Software:** For hosting, along with the finalized LMS application and database configuration.
- **Personnel:** Developers for system deployment, and support staff for post-deployment assistance.
- **Materials:** User guides, troubleshooting manuals, and video tutorials.

Site-Specific Requirements

- Reliable internet connectivity at deployment and user sites.
- A secure and scalable hosting environment to support user traffic.

By adhering to this implementation plan, the LMS will be successfully published and operational by **February 2025**, providing ABE students, Instructors, and graduates with an effective and accessible educational platform.

CONCLUSION

The development of the Learning Management System (LMS) for Agricultural and Biosystems Engineering (ABE) students, instructors, and graduates has been successfully guided by established educational and software standards. The system addresses the identified challenges in studying the Philippine Agricultural Engineering Standards (PAES) by providing a user-friendly platform with features that enhance accessibility and engagement. Through iterative development, and testing, the LMS has been optimized to meet the needs of its users while ensuring compliance with educational and data privacy regulations.

Based on the findings, the first conclusion is that the LMS effectively supports the learning of PAES through its modular design and tools, addressing the key challenges faced by ABE students, such as lack of resources, complex materials, and limited study time. Second, the LMS has been well-received during usability testing, with positive feedback highlighting its accessibility, and progress tracking, as critical features. Third, the system's adherence to the Data Privacy Act of the Philippines ensures that user data is protected, making it a secure platform for educational purposes.

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Appendix A. RESOURCE PERSONS

The researchers wish to acknowledge and extend their genuine appreciation to the contribution of the following:

Rocelle U. Camposagrado, **MIT**

Capstone Project Coordinator

Fredo T. Jalayajay, **LPT**

Capstone Project Adviser

APPENDIX B. EVALUATION TOOL/TEST DOCUMENTS

Statistical treatment of data (User Evaluation)

To evaluate the level of acceptability of the Development of a Website Learning Management System for Philippine Agricultural Engineering Standards, the proponents utilized a modified 5-point questionnaire checklist. The table below presents a 5-point scale, with 5 representing the highest level of acceptability and 1 indicating the lowest. This scale was designed to gather stakeholders' perceptions and feedback regarding the system's usability, functionality, design, and user satisfaction."

Scale	Range	Verbal Interpretation
5	4.20 - 5.0	**Strongly Agree**
4	3.40 - 4.19	**Agree**
3	2.60 - 3.39	**Neutral**
2	1.80 - 2.59	**Disagree**
1	1.00 - 1.79	**Strongly Disagree**

Statistical treatment of data (Technical Evaluation)

To evaluate the level of acceptability of the Development of a Website Learning Management System for Philippine Agricultural Engineering Standards, the proponents utilized a modified 5-point questionnaire checklist. The table below presents a 5-point scale, with 5 representing the highest level of acceptability and 1 indicating the lowest. This scale was designed to gather stakeholders' perceptions and feedback regarding the system's usability, functionality, design, and user satisfaction."

Scale	Range	Verbal Interpretation
5	2.5 - 3.00	**Strongly Agree**
4	2.00 - 2.49	**Agree**
3	1.5 - 1.99	**Neutral**
2	1.00 - 1.49	**Disagree**
1	0.5 - 0.99	**Strongly Disagree**

User's Evaluation Result

Usability	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website is easy to use.			1	12	42	**4.8**	**SA**
2. It's easy to find what I'm looking for.			1	27	27	**4.5**	**SA**
3.The menus and buttons are easy to understand.			1	7	47	**4.9**	**SA**

Design	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website looks good.			6	28	21	3.86	A
2. The text and colors are easy to read.			6	20	29	4.29	SA
3.The pages look clean and organized			4	12	39	4.33	SA

Accessibility	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website works well on my device.		1	10	20	24	4.09	A
2. The website loads quickly.			**5**	28	22	4.06	A
3.I can use the website on different devices (phone, tablet, computer).			**7**	12	36	4.23	SA

Functionality	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website features work as expected		3	9	20	23	4.15	A
2.The website allows me to complete tasks efficiently.			6	30	19	4.07	A
3.All interactive elements (e.g., buttons, links) function properly.			6	26	24	4.20	SA

Functionality	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website features work as expected		3	9	20	23	4.15	A
2.The website allows me to complete tasks efficiently.			6	30	19	4.07	A
4.The website is free from errors or glitches during use.			6	29	20	4.11	A
5.I am able to easily access all the resources and tools I need on the website.		1	4	23	27	4.38	SA

Overall	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.I am satisfied with my overall experience using the website.		2	5	31	16	4.18	A
2.I would use this website again in the future.			8	17	28	3.98	A
3.I would recommend this website to others.			4	13	36	4.42	SA

GENERAL WEIGHTED MEAN	4.28	Strongly Agree (SA)

Technical Evaluation Result

Functionality	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website's features work as expected.				1	2	2.8	SA
2. The website provides a seamless user experience.				1	2	2.8	SA
3.Navigation on the website is intuitive and logical				1	2	2.8	SA
4.Error messages (if any) are clear and informative				1	2	2.8	SA
5.All links and interactive elements function correctly.				1	2	2.8	SA

Design and Usability	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website design is visually appealing				1	2	2.8	SA
2.The website is responsive across different devices and browsers					3	3.00	SA
3.The layout supports easy navigation and accessibility					3	3.00	SA
4.The typography and color scheme enhance readability.					3	3.00	SA
5.The website adheres to modern web design standards.				1	2	2.8	SA

Performance and Security	**1**	**2**	**3**	**4**	**5**	**WM**	**VI**
1.The website loads quickly across various devices.					3	3.00	SA
2.The website's performance is stable under different loads.					3	3.00	SA
3.The layout supports easy navigation and accessibility				2	1	2.6	SA
4.Error messages (if any) are clear and informative				1	2	2.8	SA
5.The website provides clear privacy and security policies.				1	2	2.8	SA

GENERAL WEIGHTED MEAN	2.78	Strongly Agree (SA)

Instructions: Please evaluate the following aspects of Paesreview.com based on your experience.
SA-Strongly Agree | A-Agree | N-Neutral | D-Disagree | SD - Strongly Disagree

Name: Aryan Ped Sarmiento **Date: 11/25/24**

Functionality	**1**	**2**	**3**	**4**	**5**
1.The website's features work as expected.					✓
2. The website provides a seamless user experience.					✓
3.The website's security features meet industry standards					✓
4.Error messages (if any) are clear and informative					✓
5.All links and interactive elements function correctly.					✓

Design and Usability	**1**	**2**	**3**	**4**	**5**
1.The website design is visually appealing					✓
2.The website is responsive across different devices and browsers					✓
3.The layout supports easy navigation and accessibility					✓
4.The typography and color scheme enhance readability.					✓
5.The website adheres to modern web design standards.					✓

Performance and Security	**1**	**2**	**3**	**4**	**5**
1.The website loads quickly across various devices.					✓
2.The website's performance is stable under different loads.					✓
3.The website's security features meet industry standards				✓	
4.User data appears to be handled securely and confidentially.				✓	

5.The website provides clear privacy and security policies.					✓

Instructions: Please evaluate the following aspects of PAESReview.com based on your experience.
SA-Strongly Agree | A-Agree | N-Neutral | D-Disagree | SD - Strongly Disagree

Name: Mandrake Balete **Date: 11/25/24**

Website Functionality	**1**	**2**	**3**	**4**	**5**
1.The website's features work as expected.				✓	
2. The website provides a seamless user experience.				✓	
3.Navigation on the website is intuitive and logical				✓	
4.Error messages (if any) are clear and informative				✓	
5.All links and interactive elements function correctly.				✓	

Design and Usability	**1**	**2**	**3**	**4**	**5**
1.The website design is visually appealing				✓	
2.The website is responsive across different devices and browsers					✓
3.The layout supports easy navigation and accessibility					✓
4.The typography and color scheme enhance readability.					✓
5.The website adheres to modern web design standards.				✓	

Performance and Security	**1**	**2**	**3**	**4**	**5**
1.The website loads quickly across various devices.					✓

2.The website's performance is stable under different loads.					✓
3.The website's security features meet industry standards				✓	
4.User data appears to be handled securely and confidentially.					✓
5.The website provides clear privacy and security policies.					✓

Instructions: Please evaluate the following aspects of PAESReview.com based on your experience.
SA-Strongly Agree | A-Agree | N-Neutral | D-Disagree | SD - Strongly Disagree

Name: Charlee Mauricio **Date: 11/25/24**

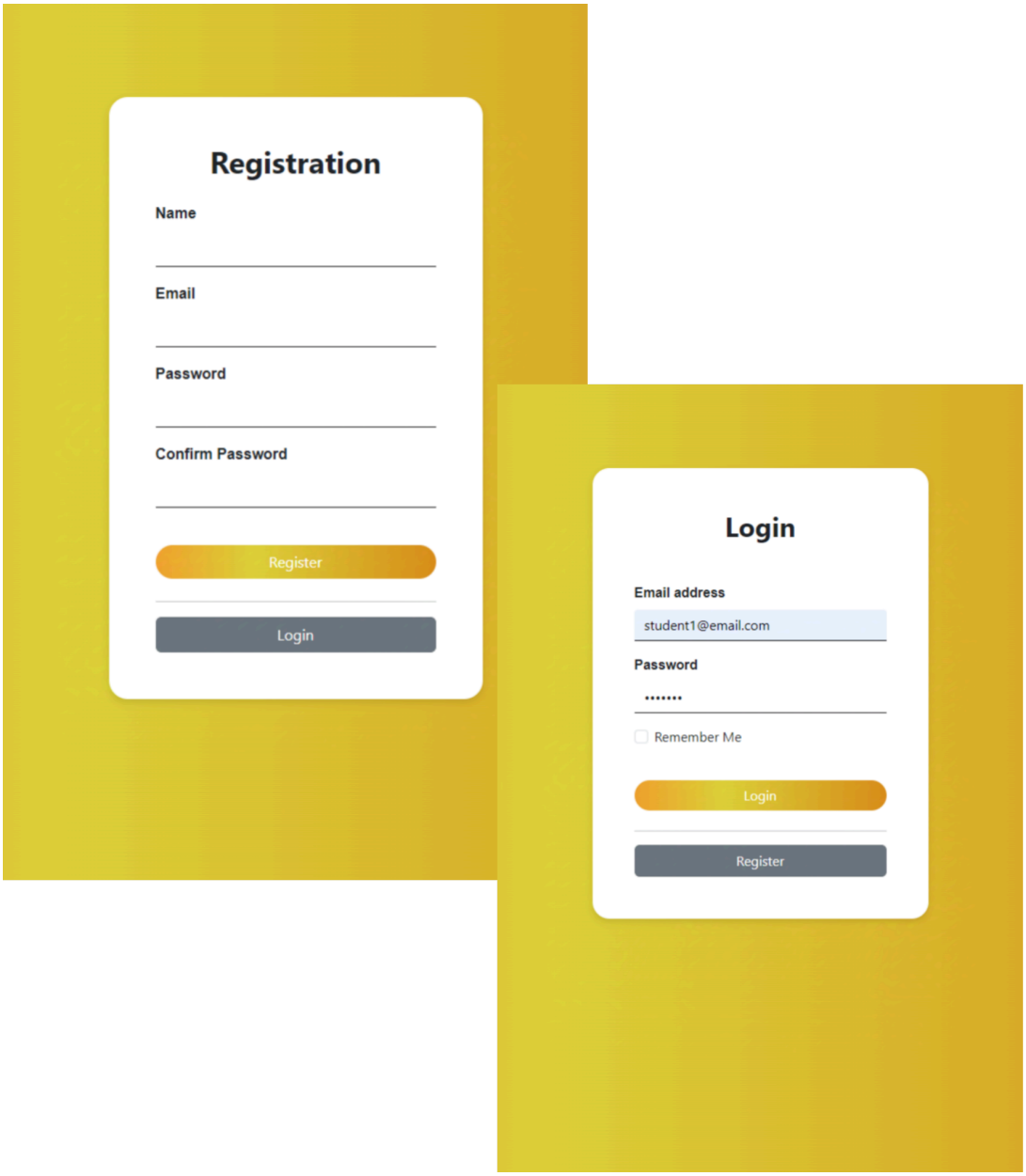
Website Functionality	**1**	**2**	**3**	**4**	**5**
1.The website's features work as expected.					✓
2. The website provides a seamless user experience.					✓
3.Navigation on the website is intuitive and logical					✓
4.Error messages (if any) are clear and informative					✓
5.All links and interactive elements function correctly.					✓

Design and Usability	**1**	**2**	**3**	**4**	**5**
1.The website design is visually appealing					✓
2.The website is responsive across different devices and browsers					✓
3.The layout supports easy navigation and accessibility					✓
4.The typography and color scheme enhance readability.					✓
5.The website adheres to modern web design standards.					✓

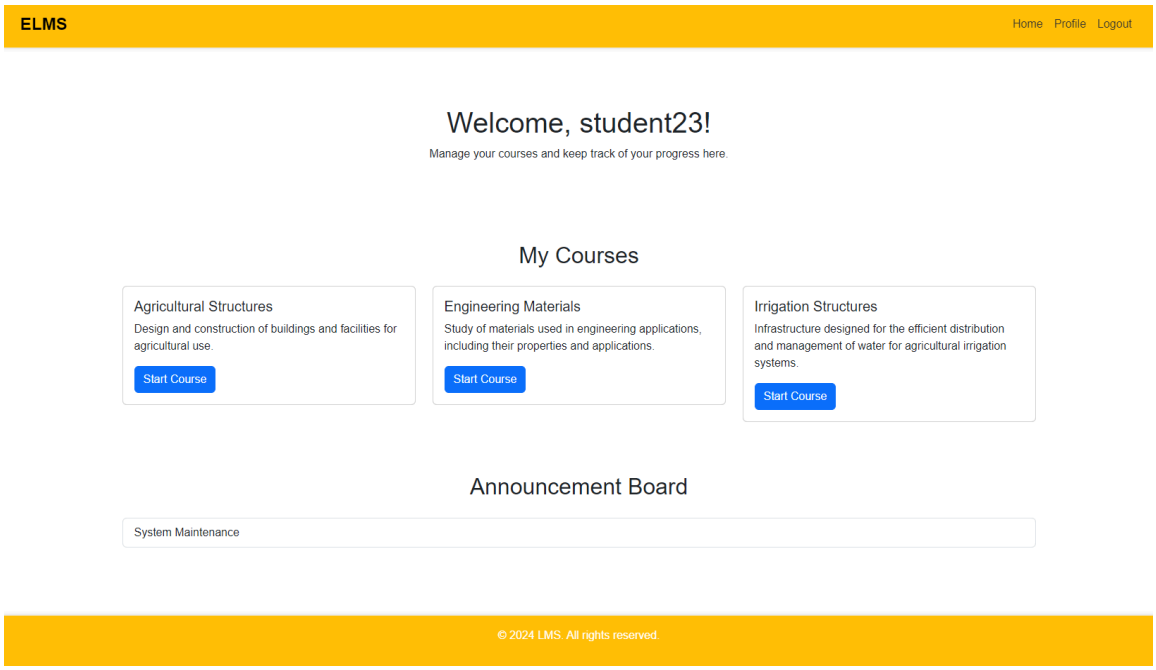
Performance and Security	**1**	**2**	**3**	**4**	**5**
1.The website loads quickly across various devices.					✓
2.The website's performance is stable under different loads.					✓
3.The website's security features meet industry standards					✓
4.User data appears to be handled securely and confidentially.					✓
5.The website provides clear privacy and security policies.					✓

APPENDIX C. SAMPLE INPUT/OUTPUT/REPORTS

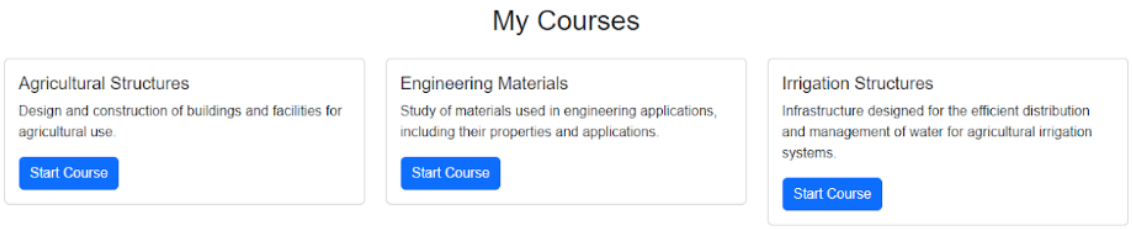
Input: Figure 1 “Registration & Login”



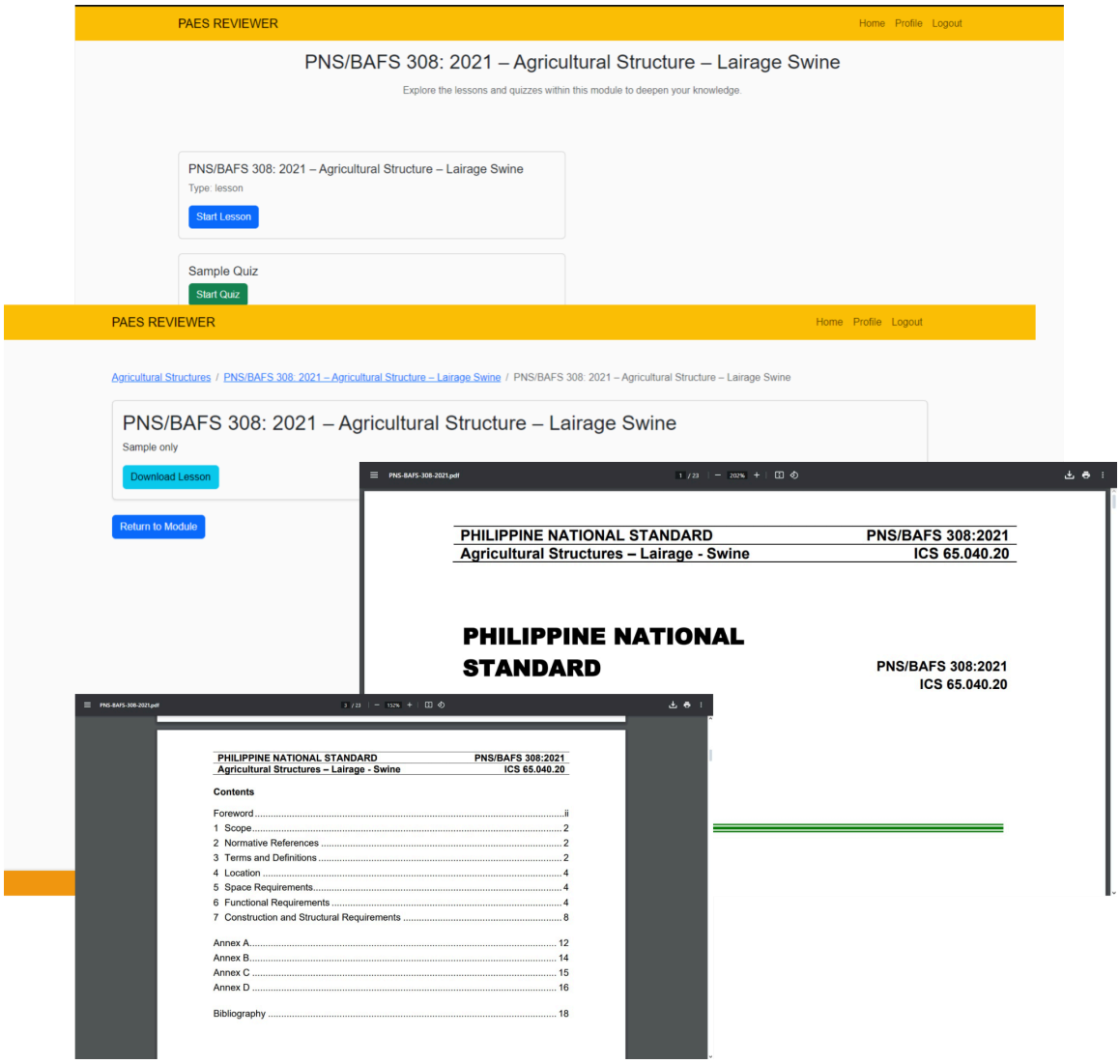
Output: Figure 1 ELMS “Dashboard”



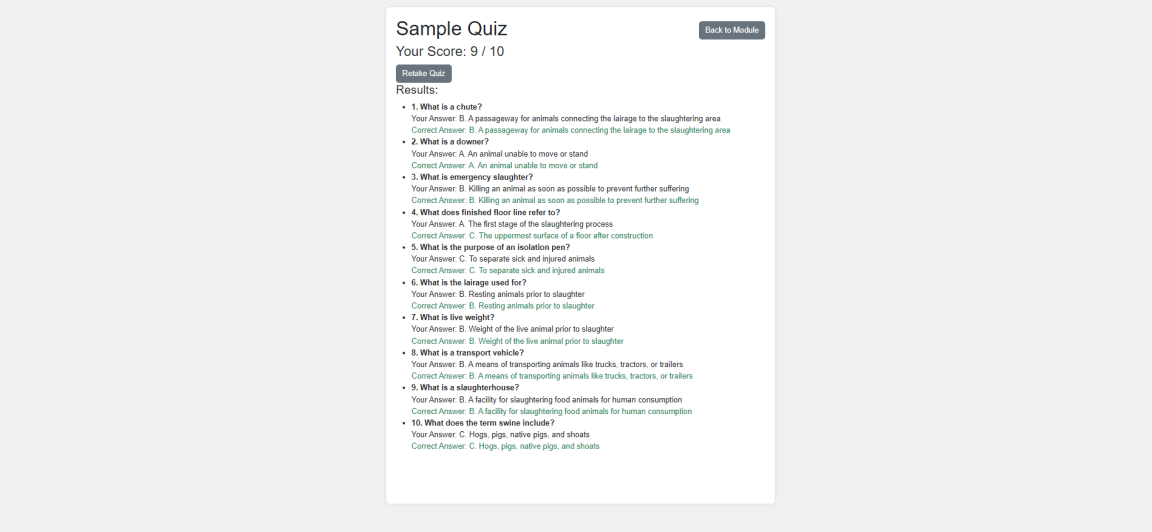
Input: Figure 2 “Courses Enrolled”



Input: Figure 3 “Courses”



Output: Figure 2 “Quiz Results”



Input: Figure 4 “Admin Dashboard”

